Australian National University
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On Laterally Confined Kelvin-Helmholtz Instabilities

Advanced Studies Course Report
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Summer Session 2024

1 Abstract

In this paper, we expand upon the standard one dimensional Kelvin-Helmholtz instability (KHI) analysis by considering the case in which the two fluids are laterally confined on an inclined plane whilst also including the viscous stabilising force, along with requiring a finite depth of the bottom fluid. To complement the presented analytical model, laboratory experiments were conducted to confirm the validity of our model. By imposing these additional boundary conditions, we are able to model conditions analogous to dense ice cascading beneath a less dense layer along an incline as observed in Alpine glacier slopes and the consequential mixing between the fluids [5]. Our results demonstrate that even including additional considerations to the base model in [1], the wave behaviour still maintains similar characteristics. Specifically, the viscous stabilising force has no effect at higher wavenumbers and as such, the force due to surface tension is reintroduced into the model so that we can rectify a monotonically increasing growth rate with respect to the along-flow wavenumber, k. In the process, we also present a novel computational technique for determining the surface tension between two layers of different salinity and as such produce a linear plot of surface tension as a function of the density of the bottom (more dense) fluid.

2 Introduction

It is well established [3, 8] that when a fluid of density ρ_1 rests on a fluid of density ρ_2 that moves with a flow speed U, an instability along the interface at a height η, may arise due to the shear following from Bernoulli's theorem - this is the Kelvin-Helmholtz instability (KHI). This phenomena can be seen in various astrophysical processes, cloud formations and mixing in ocean currents and as such, its study is of significance [8, 6]. Indeed, when we consider a linear instability analysis of the KHI under the influence of gravity, g, and surface tension, T, the following condition is necessary for an instability to form [1],

$$\frac{\rho_1\rho_2}{\rho_1+\rho_2}k^2U^2 > |k^3|T + (\rho_2-\rho_1)g|k| \tag{1}$$

with k the wavenumber. From (1) it is clear that only intermediate wavenumbers give rise to an instability and at high wavenumbers, the surface tension dominates thereby stabilising the system. However, this condition neglects any consideration of the viscous stablising forces and further assumes that both fluids are of an infinite depth. This paper aims to determine the condition for an instability to arise under these additional assumptions by determining a dispersion relation and making use of experimental data to assess the model's validity. With global warming being a prominent issue, our experiment reflects what would happen to an ice sheet, for instance, breaking off a glacier and then mixing with ocean water through the KHI. Consequently, we are able to better understand the effect of this on ocean currents, and the role of the KHI in ocean mixing of layers with a salinity difference [6]. We will use the Navier-Stokes equations to write down the governing equations and the corresponding aforementioned boundary conditions from which we seek a solution for the vertical velocities of the form $e^{i(kx+ly-\omega t)}$. Our experiment involves varying both the density, ρ_2, and flow speed, U, of the bottom fluid that flows down a laterally confined ramp at an angle θ. A camera is used to record each condition so that FIJI, an image processing software [2], can be used to determine the wavelengths of the instabilities, allowing us to compare our model to experimental data.

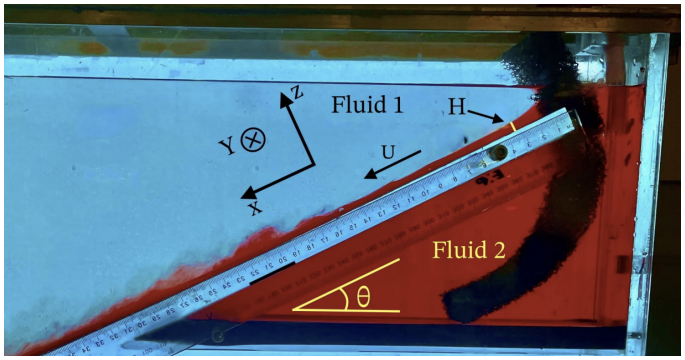


Figure 1: Schematic of the ramp where the bottom fluid (red) flows down at a speed U, leading to KHI at the interface

3 Analytical Model

Our aim in this section is to first establish the appropriate boundary conditions, allowing us to impose them on our momentum equations. By substituting an aforementioned plane wave solution, we will be able to extract our dispersion whose validity can easily be checked by setting our additional parameters to zero and inspecting if the resultant dispersion relation gives the same condition for an instability as outlined in (1).

3.1 Boundary Conditions

The first condition we consider is that at the interface between our two fluids, we require a fluid element on the

interface to move with it which we state mathematically as,

$$w_1 = w_2 = \frac{D\eta}{Dt}$$

with w_1, w_2 the vertical velocities of the top and bottom fluid, respectively. We will also require that the bottom fluid is of a finite depth, H, as can be seen in Figure 1; this will be done making use of hyperbolic functions. By including these functions, we can guarantee that w_2 will be zero at $z = -H$, i.e. we will satisfy the no-slip boundary condition. In general the top layer will be of a depth much greater then the bottom layer, and so a infinite depth assumption for the top layer is made. Given our lateral confinement, we also require a no slip in the y direction; however, experimental observations demonstrated that including this in the model was an unnecessary complication as the wave profile did not vary in y with only very minimal edge effects leading to y partials to be neglected. Further, at the interface, $z = \eta$, the pressure between the two fluids must be continuous, which is formulated as,

$$P_2 - P_1 = 0$$

where P_2, P_1 are the pressures for their respective layers. Here we choose to not include the effect of a surface tension, T, given that the interface is between two fluids of very similar properties - just a small difference in density, unlike an oil-water interface for instance and so we expect it to have negligible effects. Nevertheless, this assumption is revisited later in our paper.

3.2 The dispersion relation

We may now begin the computation for our dispersion relation by writing down the Navier-Stokes equations to model our experiment. For the upper fluid layer (fresh water) of density ρ_1 we have the following assuming an incompressible fluid with viscosity μ,

$$\rho_1 \frac{Du_1}{Dt} = -\frac{\partial P_1}{\partial x} + \rho_1 g \sin\theta + \mu\left(\frac{\partial^2 u_1}{\partial x^2} + \frac{\partial^2 u_1}{\partial z^2}\right) \quad (2)$$

$$\rho_1 \frac{Dv_1}{Dt} = -\frac{\partial P_1}{\partial y} + \mu\left(\frac{\partial^2 v_1}{\partial x^2} + \frac{\partial^2 v_1}{\partial z^2}\right) \quad (3)$$

$$\rho_1 \frac{Dw_1}{Dt} = -\frac{\partial P_1}{\partial z} - \rho_1 g \cos\theta + \mu\left(\frac{\partial^2 w_1}{\partial x^2} + \frac{\partial^2 w_1}{\partial z^2}\right) \quad (4)$$

$$\nabla \cdot \mathbf{u_1} = 0 \quad (5)$$

Where $\frac{D}{Dt} \simeq \frac{\partial}{\partial t}$ is the linearisation of the material derivative. Here we claim that any partial along the y coordinate axis is zero by the same justification we use for not considering a no slip condition in the y (cross-stream) direction - the variation along y is negligible; however, if desired the additional term would not alter the presented method so could be easily included. We then write down the governing equations for the bottom fluid for which we vary the density, ρ_2, between $(1000.1 - 1005.2) \pm 0.1\text{kg m}^{-3}$ and flows with speed U, relative to the upper fluid, down the incline.

$$\rho_2 \frac{Du_2}{Dt} = -\frac{\partial P_2}{\partial x} + \rho_2 g \sin\theta + \mu\left(\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial z^2}\right) \quad (6)$$

$$\rho_2 \frac{Dv_2}{Dt} = -\frac{\partial P_2}{\partial y} + \mu\left(\frac{\partial^2 v_2}{\partial x^2} + \frac{\partial^2 v_2}{\partial z^2}\right) \quad (7)$$

$$\rho_2 \frac{Dw_2}{Dt} = -\frac{\partial P_2}{\partial z} - \rho_2 g \cos\theta + \mu\left(\frac{\partial^2 w_2}{\partial x^2} + \frac{\partial^2 w_2}{\partial z^2}\right) \quad (8)$$

$$\nabla \cdot \mathbf{u_2} = 0 \quad (9)$$

Where $\frac{D}{Dt} \simeq \frac{\partial}{\partial t} + U\frac{\partial}{\partial x}$ is the linearisation for the material derivative in this fluid. Note that to linearise our equations we have, of course, made the approximation that $\frac{\partial}{\partial t} \gg \mathbf{u} \cdot \nabla$ which implies that ω/k (wave phase speed) is much larger than u (flow speed) which is justified given the wave amplitudes we observed we of the order 10^{-2} m. With the linearised governing equations written down, we proceed by taking an x partial of equation (2), a y partial of equation (3) and so on, yielding the following,

$$\rho_1 \frac{\partial^2 u_1}{\partial t \partial x} = -\frac{\partial^2 P_1}{\partial x^2} + \mu\left(\frac{\partial^3 u_1}{\partial x^3} + \frac{\partial^3 u_1}{\partial z^2 \partial x}\right) \quad (10)$$

$$\rho_1 \frac{\partial^2 v_1}{\partial t \partial y} = -\frac{\partial^2 P_1}{\partial y^2} + \mu\left(\frac{\partial^3 v_1}{\partial x^2 \partial y} + \frac{\partial^3 v_1}{\partial z^2 \partial y}\right) \quad (11)$$

$$\rho_1 \frac{\partial^2 w_1}{\partial t \partial z} = -\frac{\partial^2 P_1}{\partial z^2} + \mu\left(\frac{\partial^3 w_1}{\partial x^2 \partial z} + \frac{\partial^3 w_1}{\partial z^3}\right) \quad (12)$$

We can then combine equations (10) through (12) along with equation (5), so that we are left with just one, higher order, equation,

$$\rho_1 \frac{\partial}{\partial t}\left(\underbrace{\frac{\partial u_1}{\partial x} + \frac{\partial w_1}{\partial z}}_{\nabla \cdot \mathbf{u_1} = \mathbf{0}}\right) = -\nabla^2 P_1 + \mu$$
$$\left(\frac{\partial^2}{\partial x^2}\left(\underbrace{\frac{\partial u_1}{\partial x} + \frac{\partial w_1}{\partial z}}_{\nabla \cdot \mathbf{u_1} = \mathbf{0}}\right) + \frac{\partial^2}{\partial z^2}\left(\underbrace{\frac{\partial u_1}{\partial x} + \frac{\partial w_1}{\partial z}}_{\nabla \cdot \mathbf{u_1} = \mathbf{0}}\right)\right)$$

Which tells us that P_1, should have a Laplacian equal to zero. An identical procedure can be carried out for the governing equations for the bottom fluid, also resulting in the Laplacian of the pressure, P_2, equalling zero. Our no-slip boundary conditions require that w_2 approaches 0 at a depth $z = -H$, whereas w_1 approaches zero as $z \to \infty$. As such, we use these conditions, along with both pressures requiring

a wave profile in both x and y, to write down the following expressions for the pressures, up to some amplitude $\hat{P}_1$ and $\hat{P}_2$,

$$P_1 = \hat{P}_1 e^{-z\sqrt{k^2+l^2}} e^{i(kx+ly-\omega t)} - \rho_1 g z \cos\theta$$
$$P_2 = \hat{P}_2 \cosh\left(\sqrt{k^2+l^2}(z+H)\right) e^{i(kx+ly-\omega t)} - \rho_2 g z \cos\theta$$

Which evidently satisfy $\nabla^2 P_1 = \nabla^2 P_2 = 0$ along with (4) and (8) for which we include the hydrostatic pressure terms. We also want to ensure that our other boundary condition for the vertical velocity is satisfied; namely, that the wave interface, η, moves at the vertical velocity of the fluid which we write mathematically as, $\frac{D\eta}{Dt} = w_1 = w_2$ at $z = \eta$. To proceed, it is therefore necessary to obtain expressions for the vertical velocities, $w_{1,2}$ which we obtain by substituting our pressure expressions into the vertical momentum equations, (4) and (8),

$$\rho_1 \frac{\partial w_1}{\partial t} = \hat{P}_1 \sqrt{k^2+l^2} e^{-z\sqrt{k^2+l^2}} e^{i(kx+ly-\omega t)} + \mu\left(\frac{\partial^2 w_1}{\partial x^2} + \frac{\partial^2 w_1}{\partial z^2}\right)$$

Rearranging gives,

$$\left[\rho_1 \frac{\partial}{\partial t} - \mu\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}\right)\right] w_1 = \hat{P}_1 \sqrt{k^2+l^2} e^{-z\sqrt{k^2+l^2}} e^{i(kx+ly-\omega t)}$$

So w_1 has the form,

$$w_1 = \frac{\hat{P}_1\sqrt{k^2+l^2}}{\mu(k^2-(k^2+l^2)) - \rho_1\omega i} e^{-z\sqrt{k^2+l^2}} e^{i(kx+ly-\omega t)} = \frac{\hat{P}_1\sqrt{k^2+l^2}}{-(\mu l^2 + \rho_1\omega i)} e^{-z\sqrt{k^2+l^2}} e^{i(kx+ly-\omega t)}$$

And of course, w_2 is found in an identical fashion and has the expression,

$$w_2 = -\frac{\hat{P}_2\sqrt{k^2+l^2}\sinh\left(\sqrt{k^2+l^2}(z+H)\right)}{-\mu l^2 + \rho_2 U k i - i\omega\rho_2} e^{i(kx+ly-\omega t)}$$

We are now in a position to apply the interface boundary condition, which for fluid 1 we have that $\frac{D\eta}{Dt} \simeq \frac{\partial\eta}{\partial t} = w_1$ at $z = \eta$, making it evident that η is given by,

$$\eta = \frac{\hat{P}_1\sqrt{k^2+l^2}}{i\omega(\mu l^2 + \rho_1\omega i)} e^{-\eta\sqrt{k^2+l^2}} e^{i(kx+ly-\omega t)} = \frac{\hat{P}_1\sqrt{k^2+l^2}}{i\omega(\mu l^2 + \rho_1\omega i)} e^{i(kx+ly-\omega t)}$$

Where we make the approximation that $e^{-\eta\sqrt{k^2+l^2}} \approx 1$ given experimental observations demonstrated that the wavelength is much greater then η. Of course, we can also find another expression for η by using w_2 and that we must have $\frac{D\eta}{Dt} = \frac{\partial\eta}{\partial t} + U\frac{\partial\eta}{\partial x} = w_2$ in fluid 2. From this we deduce that η can also be expressed as,

$$\eta = -\frac{\hat{P}_2\sqrt{k^2+l^2}\sinh\left(\sqrt{k^2+l^2}H\right)}{(-\mu l^2 + \rho_2 U k i - i\omega\rho_2)(Uki - \omega i)} e^{i(kx+ly-\omega t)}$$

Given η must be a fixed expression that is independent of which fluid we used to determine it, our two expressions must be equal which now allows us to directly relate $\hat{P}_1$ with $\hat{P}_2$.

$$\frac{-\hat{P}_2\sinh\left(\sqrt{k^2+l^2}H\right)}{(-\mu l^2 + \rho_2 U k i - i\omega\rho_2)(Uki - \omega i)} = \frac{\hat{P}_1}{i\omega(\mu l^2 + \rho_1\omega i)}$$
$$\implies \hat{P}_2 = -\frac{(-\mu l^2 + \rho_2 U k i - i\omega\rho_2)(Uk - \omega)}{\sinh\left(\sqrt{k^2+l^2}H\right)\omega(\mu l^2 + \rho_1\omega i)}\hat{P}_1$$

We can now apply our final boundary condition that we impose on the pressures, specifically that the pressure is required to be continuous at the boundary $z = \eta$, so that $P_2 - P_1 = 0$,

$$e^{i(kx+ly-\omega t)}\left(\hat{P}_1 - \hat{P}_2\cosh\left(\sqrt{k^2+l^2}(\eta+H)\right)\right) + (\rho_2-\rho_1)g\eta\cos\theta = 0$$

Where we again use that $e^{-\eta\sqrt{k^2+l^2}} \approx 1$ and also substitute our expression for η to obtain that,

$$e^{i(kx+ly-\omega t)}\left(\hat{P}_1 - \hat{P}_2\cosh\left(H\sqrt{k^2+l^2}\right)\right) + (\rho_2-\rho_1)g\eta\cos\theta = 0$$

Substituting our expression for $\hat{P}_2$ in terms of $\hat{P}_1$ and our η expression that is also in terms of of $\hat{P}_1$, we see the following,

$$1 + \frac{(-\mu l^2 + \rho_2 U k i - i\omega\rho_2)(Uk-\omega)}{\sinh\left(\sqrt{k^2+l^2}H\right)\omega(\mu l^2 + \rho_1\omega i)}\cosh\left(\sqrt{k^2+l^2}H\right) + (\rho_2-\rho_1)g\frac{\sqrt{k^2+l^2}}{i\omega(\mu l^2+\rho_1\omega i)}\cos\theta = 0$$

Which is now an equation in ω, allowing Mathematica to easily solve for ω, our dispersion relation,

$$\omega = \frac{kU\rho_2 A}{(\rho_1+\rho_2 A)} \pm i\left(\frac{l^2\mu(1+A)}{2(\rho_1+\rho_2 A)} + \frac{\sqrt{\left(l^2\mu + (l^2\mu - 2ikU\rho_2)A\right)^2 - D}}{2(\rho_1+\rho_2 A)}\right)$$

Where we have defined the following two expressions to allow us to write down the dispersion relation,

$$A \equiv \coth(H\sqrt{k^2+l^2})$$

and,

$$D \equiv 4(\rho_1+\rho_2 A) \left(g\sqrt{k^2+l^2}(\rho_1-\rho_2)\cos\theta + kU(il^2\mu + kU\rho_2)A\right)$$

As such, we now have the desired dispersion relation so we move to understanding it in the context of our observations after first considering the experimental method.

4 Method

In this section we present the experimental setup we used which must, of course, accurately reflect our model. As such, we used a tank that had an in-built ramp set at an incline of θ radians with a width of 8 cm and hence a cross stream wavenumber, l, of $\frac{2\pi}{0.08}$ m^{-1} as this is the fundamental mode. A cavity at the top of the ramp was filled with water of density ρ_2, and then was overflown to allow said water to follow the ramp down the incline, hence leading to the shear between the two fluid layers at the interface. The flow rate, U, and in turn the height, H, were easily altered by adjusting the valve that was connected to a water tank. To ensure the flow from the tank was consistent a constant water head fountain was used. Further, to minimise large-scale mixing between the fluid layers down the ramp that would then impact the linear instability analysis that only holds at the top of the ramp, we used a sponge boat to evenly distribute fresh water to the tank. As our method for finding the flow speed, U, involved measuring the amount of time that it took for 5 litres of water to pour out of the tank by having the fresh water, we also needed to account for this. By subtracting off the amount of fresh water that was added to the tank during the time it took for the 5 litres to pour out, we are left with the flow speed. However, we further had to divide out the cross-stream and vertical dimensions to ensure that we had the flow speed, U, purely in the x direction.

4.1 Recording Data

As the aim of the experiment is to produce data to evaluate our model with, we used a video camera to obtain footage that we later analysed using the software, FIJI [2]. A ruler was aligned with the ramp to provide a conversion from pixels to length, allowing us to record the wavelength, and hence wavenumber, of the waves before they reached the point at which the aforementioned assumptions of our model break. Similarly, the recordings were also used to easily determine the height, H, and the incline angle, θ, through basic trigonometry. A camera was also mounted to view the experiment from a birds-eye view, which gave insight into the scale of the allowed assumptions in turn allowing us to neglect variation along y, as mentioned. The procedure for turning the video recordings into useful data is as follows,

1. For each trial (different U, H and ρ_2) we took screenshots at the point in which a wave was just formed and measured the trough-trough distance with the wave previous to the new one. This ensured that all the data we used would be in the linear regime of the system so that there would be a accurate correspondence between our model.

2. The height, H of the water was also easily measured using the calibrated measuring tool.

3. The above was repeated 5 times per trial and then an average of the wavelengths was taken and converted to a wavenumber.

5 Results

To begin making sense of our model, we can first determine what new condition we must have for an instability to arise from which we can verify our model by taking out the additional parameters so that our new condition reduces to (1). We can easily write our dispersion relation in terms of its real and imaginary parts, $\omega = \Re\{\omega\} + \Im\{\omega\}i$. Then we substitute this into our exponential wave profile, we see that $e^{i(kx+ly-\omega t)} = e^{i(kx+ly-\Re\{w\}t)}e^{\Im\{w\}t}$ and so, when $\Im\{w\} > 0$ an instability will of course arise; hence, the condition is simply that $\Im\{\omega\} > 0$. To revert this back to the same setup as (1), we set $l = \mu = \theta = 0$ and $\coth(H\sqrt{k^2+l^2}) = 1$ which gives precisely the same condition (just in the case where $T = 0$). With this consistency check, it is then logical to start visualising what our model predicts; specifically we can create some plots under the same conditions that our experiment allowed to then make a comparison with our observations. The following are the parameters which remain fixed for all trials and as such act as the control variables (Table 1) and all the experimental data we required for the plots in this paper (Table 2).

ρ_1 (kg m^{-3})	999.2
l (m^{-1})	78.54
θ (rad)	0.4451
μ (kg m^{-1} s^{-1})	10^{-3}

Table 1: Fixed experimental parameters

ρ_2 (kg m^{-3})	H (cm)	U (m s^{-1})	k (m^{-1})
1005.2	0.62	0.0559	220.6
1004.0	0.60	0.0529	237.5
1003.1	0.72	0.0455	212.6
1002.0	0.87	0.0342	144.7
1001.0	1.12	0.0288	184.7
1000.1	1.49	0.0199	193.9

Table 2: Recorded experimental data

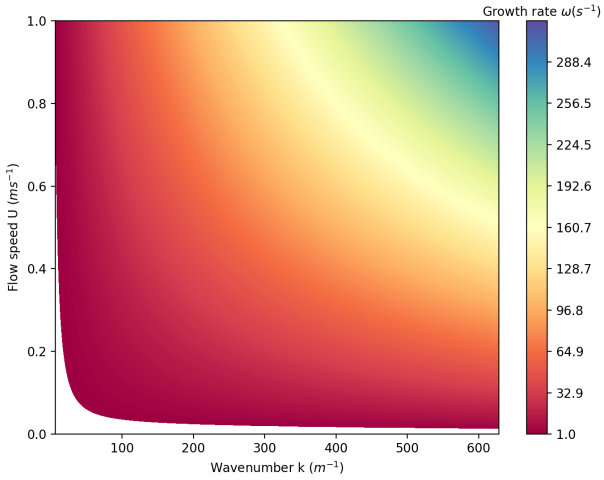


Figure 2: Contour plot for the imaginary part of the dispersion relation (growth rate) with flow speed U against wavenumber k for $\rho_2 = 1005.2$ kg m^{-3} with $T = 0$ N m^{-1}

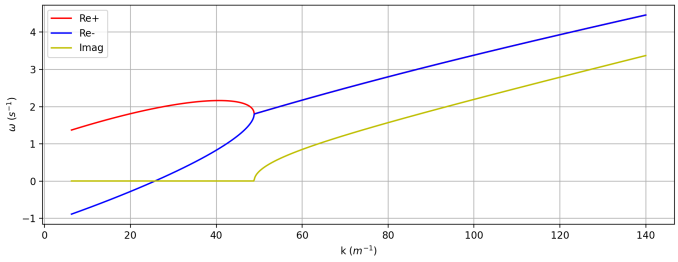


Figure 3: Growth rate of the instability (Yellow) and wave frequency (Red, Blue) against k wavenumber for $\rho_2 = 1005.2$ kg m^{-3} and $T = 0$ N m^{-1}

Figures (3)-(4) are quite interesting as they show that as the wavenumber, k, increases so too do both the growth rate, $\Im\{\omega\}$, and the wave frequency, $\Re\{\omega\}$, which contradicts the experimental values for the wavenumbers in Table 2. This is the case because only wavenumbers, k, in the interval

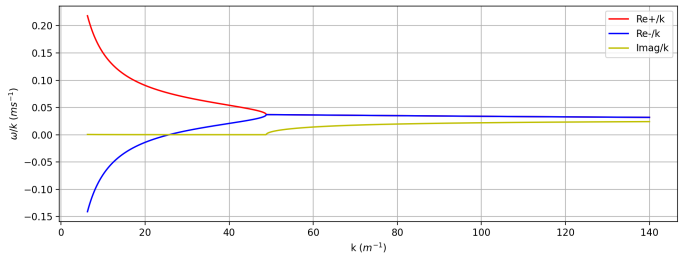


Figure 4: Wave phase speed (Red, Blue) and $\Im\{\omega\}/k$ (Yellow) against k wavenumber for $\rho_2 = 1005.2$ kg m^{-3} and $T = 0$ N m^{-1}

$144.7 < k < 237.5$ were observed to become unstable; hence, wavenumbers beyond this region must be stabilised by some force that we have not accounted for. An inspection of Table (2) shows that the average wavenumber for all trials is above 100 m^{-1} and so Figure (3) indicates that instabilities should have arisen for wavenumbers as low as 50 m^{-1} which is clearly not the case. The main implication we note from Figure (3) is that we have a growth rate that does not decrease with k and hence, we expect that for a sufficiently high wavenumber the growth rate would be some value that was not observed and as such not physically possible under the conditions of our experimental parameters. This is corroborated by Figure (2) where we see that instabilities are induced at high wavenumbers indicating a complete lack of a stabilising force for this regime. Due to these findings we now aim to solve the inconsistency within our model in the following section.

5.1 Reintroducing Surface Tension

As Figures (2)-(4) demonstrated, there appears to have been an oversight in our model, namely that viscous stabilising force has seemingly no effect at high wavenumbers implying there must be some other force that has thus far been neglected. Indeed, as we were interested in seeing if imposing conditions that had not been considered in (1) had a stabilising effect, we consequently did not consider the surface tension, T, which we can see from (1) is the dominating stabilising force at high wavenumbers for the interface between two fluids of different properties. Although we initially believed the surface tension to not be of a sufficiently large enough magnitude to contribute to the dynamics due to the rather small density difference between the layers, it seems that it may have a stabilising effect regardless. Consequently, we now work to modify our work in Section 3.2 so that the surface tension is considered. To do this, we must alter our boundary condition for the pressure; instead of requiring the pressure to be continuous at $z = \eta$, we need that the pressure is discontinuous with, $P_2 - P_1 = T\left(\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2}\right)$

[4]. This is quite a simple modification to make, as all we need to compute are the partials of η, for which we have already derived an expression for,

$$T\left(\frac{\partial^2\eta}{\partial x^2}+\frac{\partial^2\eta}{\partial y^2}\right)=-T\frac{\hat{P}_1(k^2+l^2)^{3/2}}{i\omega(\mu l^2+\rho_1\omega i)}e^{i(kx+ly-\omega t)}$$

Which we set equal to our expression for P_2-P_1 from above and again use Mathematica to obtain the following as ω,

$$\omega=\frac{-l^2\mu(1+A)+2iUk\rho_2A)}{2i(\rho_1+\rho_2A)}+\frac{\sqrt{\left(l^2\mu(1+A)-2iUk\rho_2A\right)^2-4i\left(\rho_1+\rho_2A\right)E}}{2i(\rho_1+\rho_2A)}$$

Where A is as defined above and with,

$$E\equiv i\sqrt{k^2+l^2}(g\cos\theta(\rho_1-\rho_2)-T(k^2+l^2))-AUk(l^2\mu+iUk\rho_2)$$

Although we may have reintroduced the surface tension stabilising force as desired, we lack a value to input into our model as the topic of surface tension between water layers of different salinities seems to yet have been studied in depth. As such, this needs to be calculated which we do through the following procedure,

1. Load a recording of the KHI occurring on FIJI software.
2. Calibrate length recording tool on FIJI and measure wavelength of two successive waves - only the ones at the top of the ramp are used as this is the region in which our linear analysis is applicable.
3. Record several wavelengths, take the average and convert to a wavenumber.
4. Use a Python code to increment through surface tension values until one that best fits our experimental data is determined.

The Python code looped through a large array of potential surface tension values, and for each we computed the largest growth rate, and the corresponding wavenumber. These wavenumbers were appended to a list from which the value that was closest to the experimental value, with at most 5 m $^{-1}$ difference, was chosen and the surface tension for that trial was then determined. Given that the water-air surface tension is 73 mN m^{-1} [7], we expect that our values are of a smaller magnitude given that as the density difference between the two fluids approaches zero, so too must the surface tension. The following are the results from running the code.

ρ_2 (kg m^{-3})	Surface tension (mN m^{-1})
1005.2	4.1
1004.0	3.5
1003.1	2.8
1002.0	2.4
1001.0	1.2
1000.1	0.6

Table 3: Surface Tension values between the two fluids

Which we can visualise through the following plot,

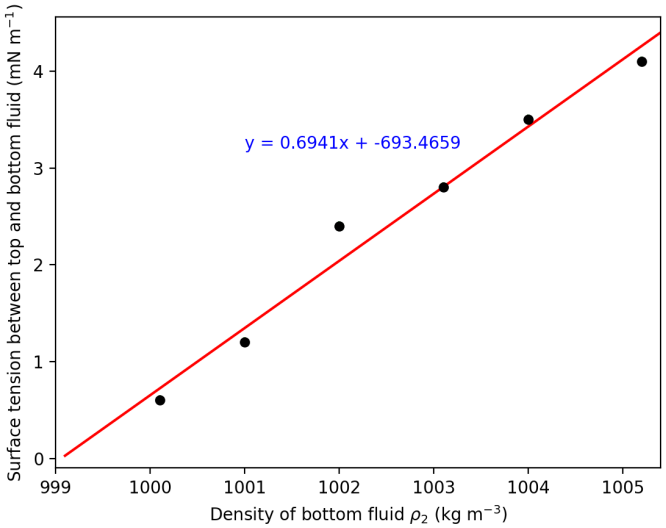


Figure 5: Surface Tension with respect to ρ_2

Indeed, Figure 5 makes it quite evident that the surface tension does behave as we hypothesised and so we can clearly see the necessity of its inclusion in our model. Note that we computed the surface tension at roughly the same flow speed, U, to guarantee that we were measuring the required relation. This is quite interesting because the relation is linear, in the given domain, but we can quickly see that outside the domain this breaks by simply evaluating the regression equation at $\rho_2=\rho_1=999.2$ kg m^{-3}. Interestingly, this does not give $T=0$ N m^{-1} but $T=0.07882$ mN m^{-1} which is not what we would expect given that they are identical fluids and so of course there cannot be any surface tension. However, given that each of our surface tension values has an associated uncertainty of 0.05 mN m^{-1}, due to how our code increments through potential surface tension values, it may be the case that this is just a computational limitation. Nevertheless, the difference between the regression and what we would expect is almost negligible. In any case, it is quite remarkable that we are able to compute such a linear relation as an indirect consequence of our model and experimental data. With our surface tension values determined, we are now able to make plots similar to Figures (2)-(4) but, of course, with the effect

of surface tension included! For the sake of brevity, we only compare the largest density difference ($\rho_2 = 1005.2$ kg m^{-3}) to the smallest ($\rho_2 = 1000.1$ kg m^{-3}) under the conditions outlined in Table (1) and (2) and the surface tension values from Table (3).

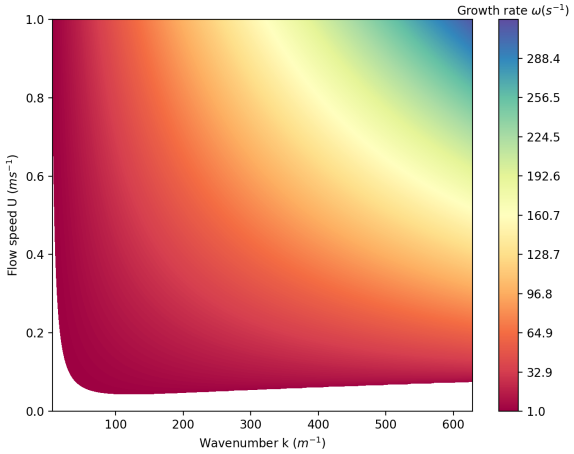


Figure 6: Contour plot for the imaginary part of the dispersion relation (growth rate) with flow speed U against wavenumber k for $\rho_2 = 1005.2$ kg m^{-3}

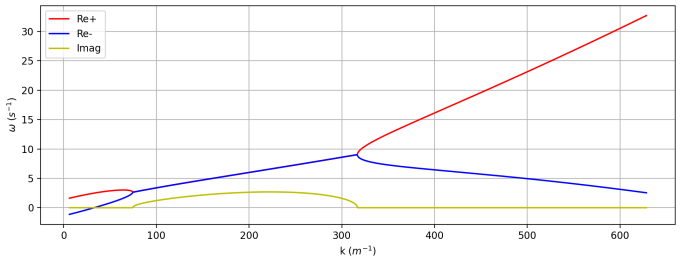


Figure 7: Growth rate of the instability (yellow) and wave frequency (Red, Blue) against k wavenumber for $\rho_2 = 1005.2$ kg m^{-3}

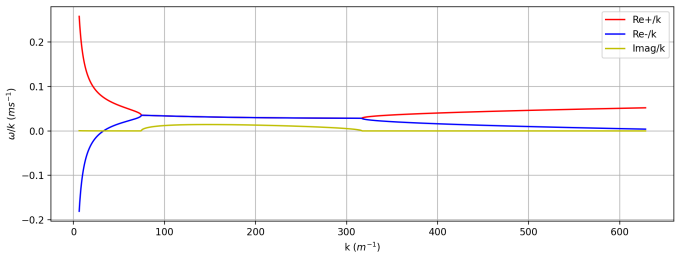


Figure 8: Wave phase speed (Red, Blue) and $\Im\{\omega\}/k$ (Yellow) against k wavenumber for $\rho_2 = 1005.2$ kg m^{-3}

Then for the smallest density difference, where $\rho_2 =$ 1000.2 kg m^{-3}

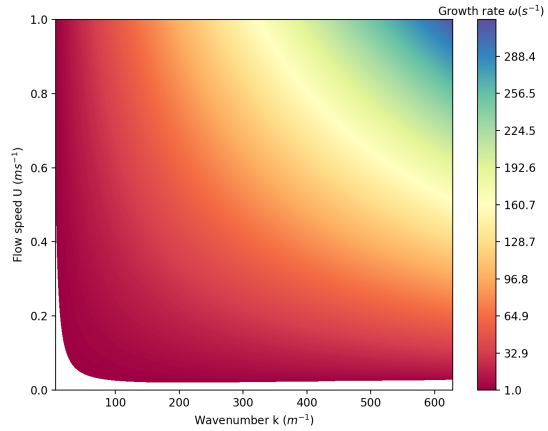


Figure 9: Contour plot for the imaginary part of the dispersion relation (growth rate) with flow speed U against wavenumber k for $\rho_2 = 1000.1$ kg m^{-3}

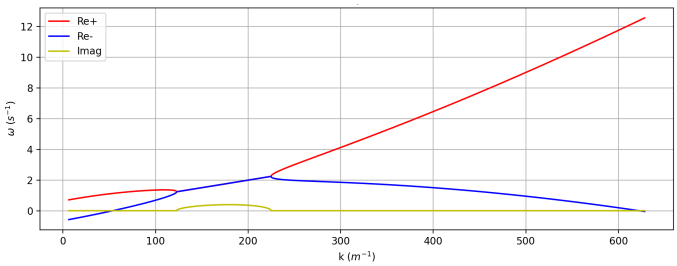


Figure 10: Growth rate of the instability (yellow) and wave frequency (Red, Blue) against k wavenumber for $\rho_2 = 1000.1$ kg m^{-3}

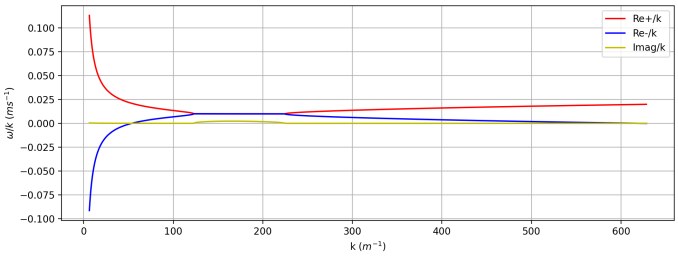


Figure 11: Wave phase speed (Red, Blue) and $\Im\{\omega\}/k$ (Yellow) against k wavenumber for $\rho_2 = 1000.1$ kg m^{-3}

5.1.1 Discussion

The main difference we see between our growth rate and phase speed plots is evidently the stabilising effect of the surface tension we reintroduced, which dominates at high wavenumbers, in exactly the same way the condition outlined in (1) stated. In fact, when we compare the domain in which $\Im\{\omega\} > 0$ in Figure (10) we can see that it is roughly $130 < k < 230$ which is much closer to the $144.7 < k < 237.5$ that was observed. Naturally, the domain from our model will be larger then what we observed because the higher growth rates were easier to observe and appeared more frequently, by definition. Even though we have considered the viscous stress, along with a cross stream

wavenumber and an incline of the plane, their effects on the system are seemingly negligible compared to the surface tension even when the magnitude is of the order of -3, the same as that of the viscosity. Indeed, this is due to the surface tension being the coefficient of the highest order (cubic) wavenumber term and the viscous terms are only linear with k, which we see in our dispersion relation. When we consider the real parts of our phase speed plots, the positive and negative solutions no longer immediately converge to $U/2$ when $\Im\{\omega\} > 0$ which happens faster for the case with $\rho_2 = 1000.1\text{kg m}^{-3}$. To see why this is the case, we must analyse the real part,

$$\begin{aligned}\Re\{\omega\}/k &= \frac{1}{k}\frac{Uk\rho_2 A}{\rho_1 + \rho_2 A} \\ &\approx \frac{U}{2} \qquad \text{(Taking } \rho_1 = \rho_2\text{)}\end{aligned}$$

And so, when ρ_1 is very close to ρ_2 our approximation holds stronger which is the case for $\rho_2 = 1000.1\text{kg m}^{-3}$ and is consequently reflected with a faster convergence to $U/2$ in our phase speed plots. Indeed, an almost identical argument holds when analysing the plot of ω just that when $\Im\{\omega\} > 0$ then $\Re\{\omega\}$ is linear with respect to k, with gradient $U/2$. We also observe that surface tension fixes the rather odd case of a non-decreasing growth rate in Figure (3) which does not make much physical sense, and is incompatible with observations, in that although high flow rates led to a faster growing instabilities, they did not grow as rapidly as suggested. Moreover, the imaginary part of our dispersion relation now has a maximal value, which corresponds to the existence of a wavenumber that has maximal growth due to the inclusion of surface tension. We see the presence of the surface tension in both the contour plots, with a greater definition in the 1005 kg m^{-3} case, where we see that at higher wavenumbers there are more flow speeds that do not induce instabilities. In any case, its evident that for lower wavenumbers, the relative scale of the contour plots is not very strong even at the higher flow speeds, indicating that the scale of the instability is not large-scale.

6 Conclusion

This paper has presented an analytical model for the case of laterally confined Kelvin-Helmholtz instabilities. Our model first considered a purely viscous stabilising force, but as a result of predictions that were inconsistent with observations, we introduced a surface tension stabilising force that in turn affirmed the existence of a maximum growth rate. Indeed, as there is no surface tension value in literature for the interface between two layers of different salinity that we have, we were able to use a Python code to find the optimal surface tension value for our model that fitted our experimental data. This allowed us to obtain a linear expression for surface tension between two layers of water with different salinities. After plotting the growth rate, wave phase speed and contour plots we saw that as anticipated the surface tension stabilises high wavenumbers which is consistent with the condition as stated in (1). However, we saw that introducing all the conditions at the scale that we had in our experiment led to a difference in results that is minimal and as such, a large scale experimental setup where the cross stream wavenumber l and depth H could be constructed to test the model in these limiting cases. Further, changing the fluids to ones of a higher viscosity would test the stabilising viscous force at a significantly higher magnitude, giving insight to its effectiveness as a stabilising force.

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